

- 9 (a) (i) number density of charge carriers/free electrons
or
number per unit volume of charge carriers/free electrons B1 [1]
- (ii) PX or QY or RZ B1 [1]
- (b) (i) V_H is inversely proportional to n B1
- for semiconductors, n is (much) smaller than for metals B1 [2]
- (ii) magnetic field would deflect holes and electrons in same direction B1
- (because) electrons are (–)ve, holes are (+)ve M1
- so V_H has opposite polarity/opposite sign A1 [3]